

Q1 Planning

- (a) Effervescence would be seen; colourless gas which forms a white ppt with $\text{Ca(OH)}_2(\text{aq})$ would be evolved.
The deep blue azurite would dissolve completely to give a blue solution.

(b) Calculation of a suitable mass of powdered rock to react with about 75% of the acid in the conical flask



Amt of H_2SO_4 in $50.00 \text{ cm}^3 = 1.00 \times 50.00 \times 10^{-3} = 0.0500 \text{ mol}$

Amt of azurite that reacts with 75% of $\text{H}_2\text{SO}_4 = \frac{1}{3} \times 0.75 \times 0.0500 = 0.0125 \text{ mol}$

Mass of azurite in $0.0125 \text{ mol} = 0.0125 \times 344.5 = 4.306 \text{ g}$

Mass of powdered rock to be used $= \frac{4.306}{0.90} = 4.78 \text{ g}$

Dilution and Volume of unreacted sulfuric acid required for titration

Amt of excess H_2SO_4 in reaction mixture $= 0.25 \times 0.0500 = 0.0125 \text{ mol}$

Assume that volume of NaOH required for titration is 25.00 cm^3

Amt of excess H_2SO_4 used for titration $= \frac{1}{2} \times 0.100 \times 25 \times 10^{-3} = 0.00125 \text{ mol}$ (ie 10% of 0.0125 mol)

Hence, 10% of excess H_2SO_4 is needed for titration with $\text{NaOH}(\text{aq})$.

So, the final reaction mixture can be diluted to 250 cm^3 and 25.0 cm^3 (ie 10% of 250 cm^3) will be pipetted for titration with NaOH .

Procedure

1. Using an electronic balance, weigh out accurately about 4.78 g of the powdered rock in a clean and dry weighing bottle. Record the mass of the weighing bottle and powdered rock.
2. Transfer the powdered rock sample into the 250 cm^3 conical flask containing 50.00 cm^3 of sulfuric acid and swirl the contents. Place a glass filter funnel on the mouth of the conical flask to prevent acid spray.
3. Reweigh the emptied weighing bottle and record its mass.
4. When effervescence has ended and all the powdered rock has dissolved, transfer the final reaction mixture quantitatively into a 250 cm^3 graduated flask with the aid of a funnel and a glass rod. Rinse the conical flask and the glass filter funnel a few times with small volumes of deionised water each time and transfer all the washings into the graduated flask.
4. Fill the graduated flask to the 250 cm^3 mark with more deionised water. Use a teat pipette (or dropper) to add the deionised water drop by drop when nearing the mark.
5. Stopper the graduated flask and shake the solution thoroughly to ensure that it is homogeneous. Label the solution **FA 3**.

6. Pipette 25.0 cm³ of **FA 3** into a 250 cm³ conical flask. Add 2 drops of phenolphthalein indicator.
7. Fill the burette with the 0.100 mol dm⁻³ NaOH(aq) provided. Titrate the solution in the conical flask with the standard dilute NaOH(aq) placed in the burette.
8. Stop the titration when one drop of the NaOH(aq) added changes the colour of the solution in the conical flask from blue to light purple.
9. Repeat the titration until at least two consistent results are obtained, i.e. the two titre volumes do not differ by more than 0.10 cm³.

Calculations

Let that the average values of two consistent titres be 1000b cm³ or b dm³.
and

mass of powdered rock sample and weighing bottle / g = d

mass of emptied weighing bottle / g = e

mass of powdered rock sample /g = d – e = c

Amt of excess sulfuric acid in 25.0 cm³ of **FA3** = $\frac{1}{2} \times b \times 0.100 = 0.0500b$ mol

Amt of excess sulfuric in reaction mixture = $0.0500b \times \frac{250}{25.0} = 0.500b$ mol

Amt of sulfuric acid reacted with azurite = 0.0500 – 0.500b mol

Amt of pure azurite present in the powdered rock sample = $\frac{1}{3} \times (0.0500 - 0.500b)$ mol

Mass of pure azurite present in the powdered rock sample = $\frac{1}{3} \times (0.0500 - 0.500b) \times 344.5$
= 114.8 x (0.0500 – 0.500b) g

Percentage by mass of pure azurite present in the powdered rock sample
= $\frac{114.8 \times (0.0500 - 0.500b)}{c} \times 100 \%$

Question 2

(a)(i) Individual fatty acids have intermolecular hydrogen bonds which are stronger than the permanent dipole-permanent dipole intermolecular forces between triester. Fatty acids have lower boiling points than trimester as more energy is needed to overcome the strong hydrogen bonding between fatty acid molecules, resulting in its higher boiling point.

(ii) stearic acid has longer carbon chain (2 more carbon atoms) than palmitic acid. As the chain length of alkyl group, R, increases, the instantaneous dipole-induced dipole forces between RCO_2H molecules become more predominant. More energy is needed to overcome these forces.

(iii) The presence of $\text{C}=\text{C}$ bonds in fatty acids with the same carbon atoms lowers the melting point of the fatty acids. [m.p. of linolenic acid < linoleic acid < oleic acid < stearic acid;]

(b)(i) Electrophilic addition

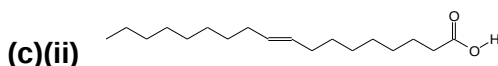
(b)(ii) Mass of iodine that would react with 100g of olive oil = $(100/0.256) \times 0.237 = 92.6 \text{ g}$

(b)(iii) Amt of iodine in 92.6 g = $92.6 / 254 = 0.3646 \text{ mol}$
Amt of $\text{C}=\text{C}$ bonds in olive oil = 0.3646 mol
Amt of olive oil in 100 g = $100/782 = 0.1279 \text{ mol}$
Average number of $\text{C}=\text{C}$ bonds = $0.3646/0.1279 = 2.85$

(b)(iv) Triesters containing 3 oleic acids contain 3 $\text{C}=\text{C}$ bonds per molecule.
As the average number of $\text{C}=\text{C}$ bond in each molecule of olive oil is 2.85 which is less than 3 and triesters of olive oil are mainly formed from oleic acid, olive oil should contain triesters of palmitic acid and/or stearic acid as the other significant component. Since the

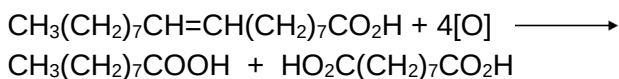
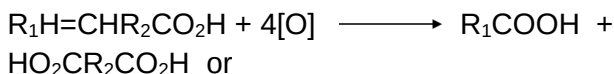
average Mr of olive oil is smaller than that of triester of oleic acid (Mr = $282 \times 3 + 12 \times 3 + 2 = 884$), the other major component of olive oil should be triester of palmitic acid which has a lower Mr than that of stearic acid.

(c)(i) H_2 , Ni catalyst, heat



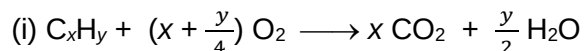
(d) ester undergoes hydrolysis

alkene($\text{C}=\text{C}$) undergoes oxidative cleavage



Question 3

(a) Silicon carbide has a giant molecular structure. Each silicon atom is covalently bonded to four other carbon atoms arranged tetrahedrally around it. Each carbon atom is also covalently bonded to four other silicon atoms. This tetrahedral arrangement is repeated throughout the whole molecule. Melting requires a lot of energy to break the strong covalent bonds between all atoms, hence silicon carbide has very high melting point.



(b) Heterogeneous catalysis, the catalyst and the reactants are in different phases.
For heterogeneous catalysis to occur, the reactant molecules need to be readily adsorbed onto the catalyst surface. The adsorption of the reactant molecules at the

catalyst surface increases the reaction rate because it

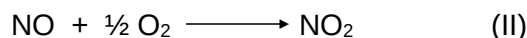
1. weakens the covalent bonds within the reactant molecules, thereby reducing the activation energy for the reaction.
2. increases the concentration of reactant molecules at the catalyst surface and allows the reactant molecules to come into close contact with proper orientation for reaction.

(c) (i) N(-3) in NH_3 to N(0) in N_2

N(+4) in NO_2 to N(0) in N_2



(d) (i) $\text{NO}_2 + \text{SO}_2 \longrightarrow \text{SO}_3 + \text{NO}$ (I)



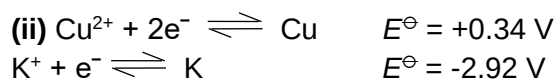
Homogeneous catalyst; oxidizes SO_2 to SO_3 , and is regenerated (II)

(ii) Cause breathing difficulties; form acid rains, corrodes buildings

(iii) $K_p = \frac{p_{\text{N}_2\text{O}_4}}{p_{\text{NO}_2}^2} = \frac{p_{\text{N}_2\text{O}_4}}{(1.5 \times 10^{-3})^2} \text{ Pa}^{-1}$
 $= 6.25 \times 10^{-5} \text{ Pa}^{-1}$
 $p_{\text{N}_2\text{O}_4} = 1.41 \times 10^{-10} \text{ Pa}$

Question 4

(a) (i) K: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$
 Cu: $1s^2 2s^2 2p^6 3s^2 3p^6 3d^9 4s^1$



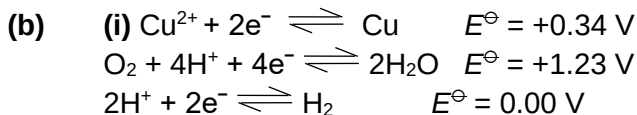
First ionisation energy of K is $+418 \text{ kJmol}^{-1}$ while that for Cu is $+745 \text{ kJmol}^{-1}$.

From both the standard reduction potential and ionisation energies, it can be seen that it is much easier for K to lose an electron than for Cu to lose an electron.

K loses an electron to attain a stable octet electronic configuration. For Cu, the energy levels of the 3d and 4s subshells are

approximately the same and Cu can exhibit variable oxidation states.

There is also a slight increase in shielding from K to Cu, causing the valence electron in Cu to be less readily removed.



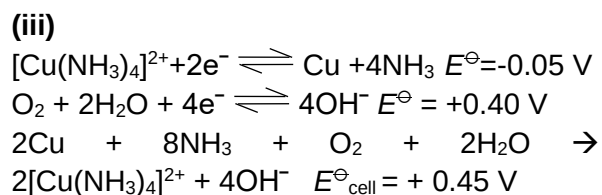
In acid without O_2 ,

$E^\ominus_{\text{cell}} = 0 - 0.34 = -0.34 \text{ V} < 0$ (not feasible)

In acid with O_2 ,

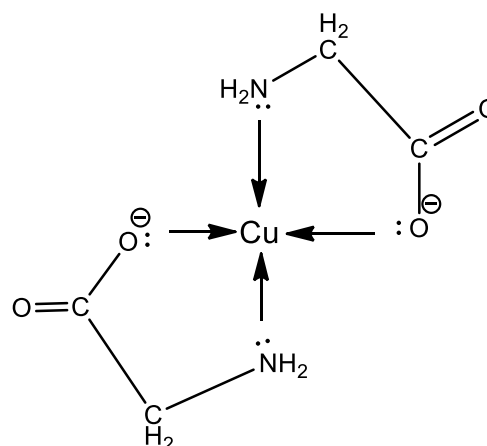
$E^\ominus_{\text{cell}} = 1.23 - 0.34 = +0.89 \text{ V} > 0$ (feasible)

(ii) Even though the reaction is thermodynamically feasible, the activation energy is high and hence the rate of reaction is slow.



(c) (i) A ligand is an ion or a molecule which contains at least one atom bearing a lone pair of electrons which can be donated into a low-lying vacant orbital of a central metal atom or ion forming a co-ordinate bond (or dative covalent bond), and resulting in the formation of a complex.

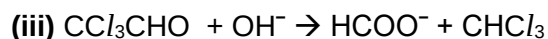
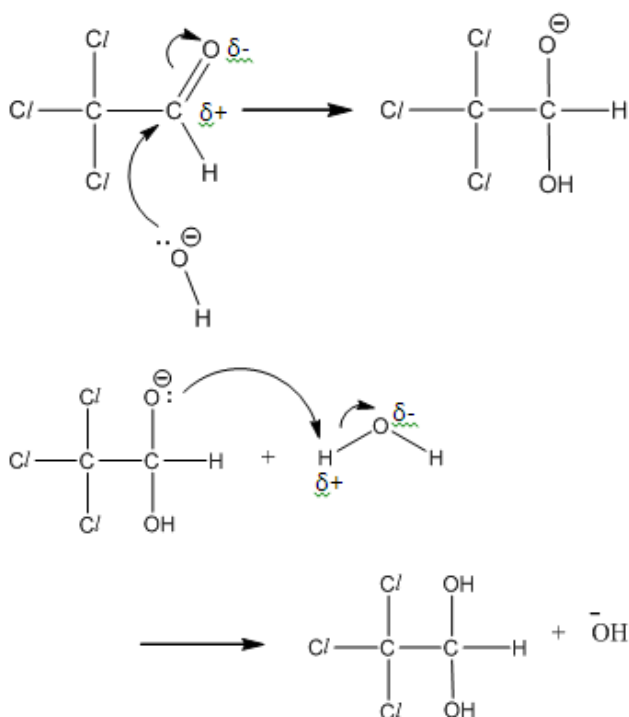
(ii)



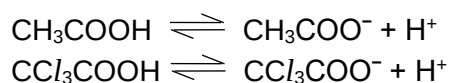
Question 5

- (a) (i) Add 2,4-dinitrophenylhydrazine to the mixture. If orange precipitate is seen, there is some remaining trichloroethanal.

(ii)

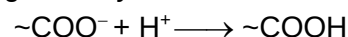


- (b) (i) The acidity of a compound depends on the relative stability of its conjugate base anion (A^-). The more stable the conjugate base, the more acidic the compound will be. The anion may be stabilized by dispersal of negative charge.



The conjugate base of trichloroethanoic acid is more stable than that of ethanoic acid as the negative charge can be dispersed over 3 electronegative Cl atoms.

(ii) Acids disrupt ionic interactions by protonating ionic side chain R groups containing carboxylate anions:



The protonated carboxyl groups are then unable to participate in ionic interactions with side chains containing -NH_3^+ , bringing about denaturation of the protein.

